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Published in:
2021 International Conference on Robotics and Automation (ICRA 2021), IEEE

DOI:
10.1109/ICRA48506.2021.9562118

Publication date:
2021

Document version:
Accepted manuscript

Citation for pulished version (APA):
Xiong, X., & Manoonpong, P. (2021). A Variable Soft Finger Exoskeleton for Quantifying Fatigue-induced Mechanical Impedance. In *2021 International Conference on Robotics and Automation (ICRA 2021), IEEE* (pp. 10347-10352). IEEE. https://doi.org/10.1109/ICRA48506.2021.9562118

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A Variable Soft Finger Exoskeleton for Quantifying Fatigue-induced Mechanical Impedance

Xiaofeng Xiong[1] and Poramate Manoonpong[1,2]

***Abstract*— Interactive (mechanical) impedance and finger fatigues are important topics, which have not been well investigated. To tackle this problem, we developed a soft lightweight (0.25 kg) finger exoskeleton (TIE-EXO) for quantifying interactive impedance and finger fatigue. A resist-as-needed (RAN) controller was used to produce variable resistance in fingers' exercises. The TIE-EXO's feedback and RAN's parameters were applied to quantify the relationship between interactive impedance and finger fatigue. This quantification was validated in the index and middle fingers of three subjects. This validation shows that the RAN control enables the TIE-EXO to produce online resistance adaptations to different subjects and finger fatigue. Moreover, it indicates a variation and invariance in finger impedance control. We argue that the proposed method provides a novel way for investigating interactive impedance and finger fatigue.**

I. Introduction

Limb mechanical impedance is the complex relationship between joint kinematics (i.e., rotational angles and velocities) and the resulting dynamics (e.g., torques) [1], [2]. It consists of stiffness defined as the ratio between a torque and angle change, and damping given by the ratio between the torque and velocity change. Numerous studies have shown that adaptive mechanical impedance facilitates versatile upper limb manipulation for physical contact tasks (e.g., grasping) [3], [4]. There are fewer mechanical impedance studies on human fingers owing to more complex anatomy and structures, compared to those on arms. Finger impedance modulation is a key for humans to skillfully interact with physical objects [1]. Fingers exhibit intrinsic spring- and damping-like properties to prevent unstable movement in interaction. Therefore, quantifying interactive impedance can shed light on fundamental principles underlying finger musculoskeletal control. However, such quantification has not yet been provided by an online impedance adaptation mechanism (exoskeleton). The main focus of this paper is to present a simple and novel mechanism for identifying and quantifying interactive impedance and finger movement. This identification and quantification were achieved by a soft lightweight (0.25 kg) finger exoskeleton (TIE-EXO) and its resist-as-needed (RAN) controller. They were utilized to produce variable resistance in fingers' exercises.

Pure soft or rigid components were adopted in most current finger exoskeleton designs, which may increase their complexities and reduce usability [5]. Moreover, some designs usually require cables that transmit actuator torques to fingers, thereby increasing sensor noise and control nonlinearity [6]. To address these problems, a soft-rigid hybrid design was utilized in the developed TIE-EXO. Soft strip bands were used to tie an actuator and direct-drive transmission mechanism to a finger. Since there is no rigid framework to fix the TIE-EXO actuator, it requires an adaptive controller for producing stable movements to prevent its slides. Therefore, a resist-as-needed (RAN) controller was developed for online impedance adaptations to finger movement. The impedance (e.g., stiffness) parameters of existing controllers were manually tuned to assist human finger movements [7], [8], [9], [10]. This is not functionally plausible, since humans can online adapt interactive (mechanical) impedance to various physical contact tasks [1], [11]. Such implausibility may cause control instability in finger and exoskeleton interaction [1]. To tackle this problem, an online impedance adaptation (i.e., RAN) controller was developed for stable interaction control. Additionally, no controller or model had been developed for quantifying fatigue-induced interactive impedance by means of an exercise exoskeleton. A review of state-of-art finger hardware and control designs is given by [5], [12], [13]. The discussion on related controllers for identifying mechanical impedance can be seen in section IV.

Taken together, the main contributions of the presented study are as follows:

- A soft lightweight (0.25 kg) finger exoskeleton design for physical exercises.
- A resist-as-needed controller for the exercise exoskeleton to online produce variable resistance against wearers' fingers.
- A simple method for quantifying fatigue-induced mechanical impedance by using the TIE-EXO and RAN.

The proposed TIE-EXO provides online impedance adaptation for a novel application (fatigue quantification), which has not yet been presented by state-of-art exoskeletons [7], [8], [9], [10]. Their comparison is given in TABLE I, while an intensive one is shown in [13]. Note that the above-mentioned contributions are excluded from our previous works aiming mainly at online adaptive exoskeleton control [14], [15], rather than mechanical impedance quantification

This research was funded by H2020-EU FET Proactive (Grant no. 732266, P.M. = Project Co-PI), Human Frontier Science Program (Grant no. RGP0002/2017, P.M. = Project Co-PI), and a startup grant on bio-inspired robotics from the Vidyasirimedhi Institute of Science and Technology (VISTEC, P.M. = Project PI).

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TABLE I

COMPARISON BETWEEN THE PROPOSED (TIE-EXO) AND STATE-OF-ART EXOSKELETONS [13].

Device name	TIE-EXO	EXO-Glove [7]	Maestro [8]	Tenoexo [9]	HX-β [10]
Year	2021	2021	2020	2020	2019
Function	Haptic (resist)	Rehabilitation	Assist	Assist	Rehabilitation
Actuation	DC motor and shaft	DC motor and cables	SEA and cables	Stepper motor, cables, springs	SEA and cables
Design	Soft-rigid hybrid	Redundant	Redundant	Soft-rigid hybrid	Redundant
Online impedance adaptation	Yes	No	No	No	No
Fatigue investigation	Yes	No	No	No	No
Portable	Yes	Yes	Yes	Yes	No
DOF	1	3	2	3	5
Weight [kg]	0.25	0.863	0.08	0.64	0.42
Max. Torque [$N \cdot m$]	4.1	1.498	0.12	0.2	0.5

presented in this study. Moreover, the RAN exhibits online resistance adaptations to different wearers in this study.

The remainder of this paper is structured as follows. The TIE-EXO design and RAN controller are presented in section II. Section III contains the experimental results of quantifying fatigue-induced mechanical impedance of fingers' movements using the TIE-EXO and RAN. A discussion and conclusion are provided in section IV.

II. HARDWARE AND SOFTWARE DESIGNS

A. Soft lightweight finger exoskeleton (TIE-EXO)

The TIE-EXO consists of a soft glove, actuator, rigid shaft, USB communication interface, power converter, and three soft strip bands (see Fig. 1).

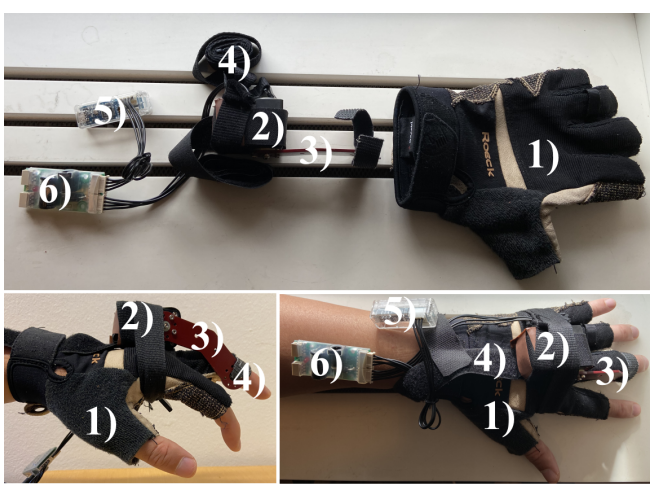

Fig. 1. Soft lightweight (0.25 kg) finger exoskeleton (TIE-EXO). 1) Glove. 2) Actuator. 3) Rigid shaft. 4) Soft strip bands. 5) USB communication interface. 6) Power converter. Row 1 illustrates the components of the TIE-EXO, while row 2 depicts the side and top views of the TIE-EXO worn on a hand.

The design of the TIE-EXO complies to the following principles presented in [5]:

- **Hand anatomy**. The wearable design allows the TIE-EXO to be worn on different fingers. A soft strip band was used to tie a middle phalanx to the rigid curved shaft (see 3 and 4 in Fig. 1). The shaft acts as the force transmission unit for the actuator (see 2 in Fig. 1).
- **Safety**. A resist-as-needed (RAN) controller (see Fig. 4) was developed to generate variable soft motions for different wearers.
- **Comfort**. The soft glove and strip bands were used to tie and fix the actuator to a hand (see 1 and 4 in Fig. 1), rather than a rigid mechanical framework. The actuator is lightweight (0.082 kg, see 2 in Fig. 1). Soft components and a lightweight actuator reduce weight and increase portability, thereby maximizing wearers' comfort and usability.
- **Effective force transmission**. The rigid shaft was selected as the force transmission unit (see 3 in Fig. 1). Its simple direct-drive and curved design reduces mechanic and control complexities of the TIE-EXO.
- **Affordability**. The hardware cost of the TIE-EXO is less than 500 EUR (see its specification in TABLE II). Its actuator was selected in terms of higher affordability and torque/weight ratio, compared to others. The selection criteria can be seen in [15].

These advantages were justified through the experiments of quantifying interactive impedance, using the TIE-EXO and its RAN controller. For instance, the RAN allows the TIE-EXO to produce variable smooth resistance against fingers, thereby guaranteeing quantification safety.

TABLE II

TIE-EXO SPECIFICATION

Weight	Actuator	Stall torque	Voltage
0.25 [kg]	Dynamixel-XM430	4.1 [$N \cdot m$]	12.0 [V]

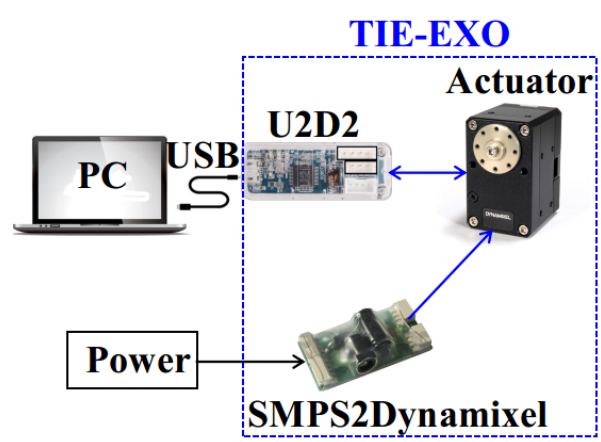

Fig. 2. TIE-EXO communication interface and power converter.

SMPSDynamixel acts as a power converter, while U2D2 serves as a USB communication interface between a PC and

the actuator (see Fig. 2). The U2D2 interface allows the RAN controller implemented on the PC to send actuator commands and receive joint feedback (i.e., position and velocity).

The inverse dynamics model of the TIE-EXO and its wearer (see Fig. 3) is given by [16],

$$I\ddot{q}+H(\dot{q})+G(q)=\tau+\tau_h, \tag{1}$$

where I, H, and G represent the inertial, viscous, and gravity terms determined by the exoskeleton acceleration $\ddot{q}$, velocity $\dot{q}$, and position q. Note that all dynamic parameters (e.g., I) of Eq. (1) and the human torque τ_h are unknown. The joint resistance torque τ is generated by the proposed RAN controller, in which the unknown dynamics are integrated to optimize the impedance gains $k_p(t)$ and $k_d(t)$ (see Eq. 4).

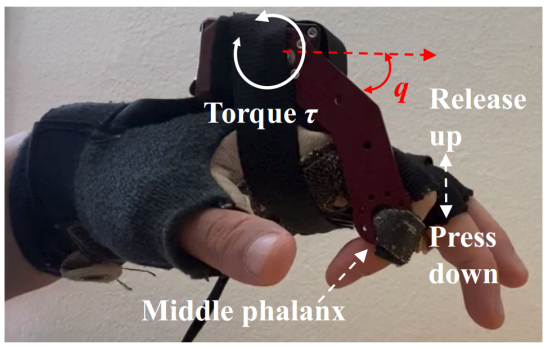

Fig. 3. Inverse dynamics model of the TIE-EXO and its wearer (see Eq. (1)). A soft strip band is used to tie a middle phalanx to a rigid curved shaft (see 3) and 4) in Fig. 1).

B. Resist-as-needed (RAN) Controller

The RAN controller is developed to generate the joint resistance torque τ based on a proportional-derivative (PD) rule [17] (see Fig. 4),

$$\tau=-k_p(t)e(t)-k_d(t)\dot{e}(t), \tag{2}$$

where $k_p(t)$ and $k_d(t)$ denote the impedance gains. $e(t)$ and $\dot{e}(t)$ represent the joint position and velocity errors given by [17],

$$\begin{aligned} &e(t)=q(t)-q_d(t), \dot{e}(t)=\dot{q}(t)-\dot{q}_d(t), \\ &\varepsilon(t)=e(t)+\beta\dot{e}(t), \beta=0.05, \end{aligned} \tag{3}$$

where $\varepsilon(t)$ denotes the joint tracking error.

The RAN controller aims at co-minimizing the impedance efforts and motion errors of the TIE-EXO (see Fig. 3) over the time period T using the following objective function (simplified from [18]),

$$\begin{aligned} J_o(t)&=J_c(t)+J_p(t), \\ J_c(t)&=\frac{1}{2}\int_{t-T}^{t}(k_p(t))^2+(k_d(t))^2, \\ J_p(t)&=\frac{1}{2}\int_{t-T}^{t}I(\varepsilon(t))^2, \end{aligned} \tag{4}$$

where J_o is the overall objective function consisting of the impedance effort J_c and tracking task J_p costs, with respect to the TIE-EXO dynamics and control models (see Eqs. (1) and (2)). The RAN co-minimization leads to the exoskeleton joint impedance adaptation given by (see Fig. 4),

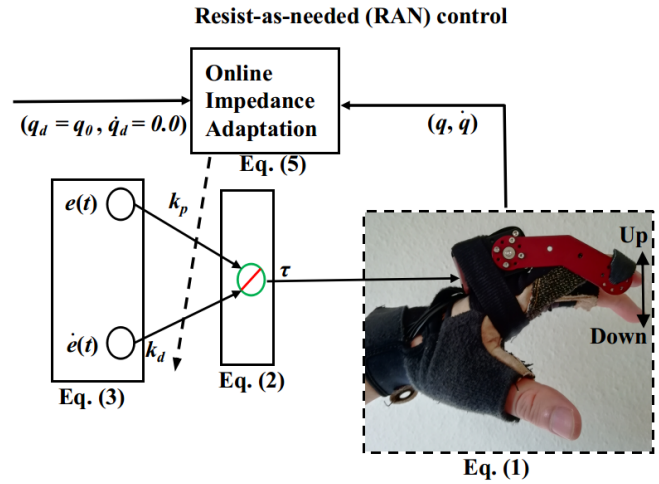

Fig. 4. Resist-as-needed (RAN) control for online mechanical impedance adaptation.

$$\begin{aligned} &k_p(t)=f(t)e(t), k_d(t)=f(t)\dot{e}(t), \\ &f(t)=\frac{\varepsilon(t)}{\gamma(t)}, \gamma(t)=\frac{a}{1+b\varepsilon(t)^2}, \end{aligned} \tag{5}$$

where $e(t)$ and $\dot{e}(t)$ denote the position and velocity errors (see Eq. 3), while $\gamma(t)$ is an adaptation scalar with the positive scalars $a=0.2$ and $b=5$. The values of a and b were chosen to adapt the response speed. All scalars, as well as the derivation of Eqs. (4) and (5) refer to our developed human-like impedance controller [18]. The stability proof of the online impedance adaptation (i.e., Eq. (5)) can be seen in [18], [17]. Online resistance adaptation is achieved by setting the desired joint position and velocity $(q_d, \dot{q}_d)$ to the initial states, i.e., $q_d=q_0, \dot{q}_d=0.0$ (see Fig. 4). The generated joint resistance torque τ increases with the increment of joint position e and velocity $\dot{e}$ errors (see Eq.(5)). Note that the online impedance adaptation mechanism has been successfully applied to our elbow and finger exoskeletons [14], [15], respectively. However, the previous works focus on adaptive impedance control, rather than quantifying fatigue-induced interactive (mechanical) impedance presented in this paper.

III. Experiments

The RAN controller was run to control the TIE-EXO worn on the fingers of three healthy subjects who are two males (i.e., S1 and S2) and a female (i.e., S3). An experimental video link is shown in [19]. A soft strip band was used to tie a middle phalanx to a rigid curved shaft (see Fig. 3). The subjects were asked to move index and middle fingers, while the movement (i.e., down and up) results in variable resistant torques generated by the RAN controller. The subjects felt uncomfortable, if they were asked to continually performed more than 10 trials on a finger. Therefore, each subject continually performed 10 trials for each finger, and each trial was run 15.0 s. There was a 15-minute break between the two series of trials on index and middle fingers. The relationship

between finger fatigue Q and average mechanical impedance $(\overline{k_p}, \overline{k_d})$ is quantified by,

$$Q(\overline{q}, \overline{\dot{q}}) = Q(f_1(\overline{k_p}), f_2(\overline{k_d})), \tag{6}$$

where the fatigue function Q is quantified by the average movement angle $\overline{q}$ and velocity $\overline{\dot{q}}$ of the TIE-EXO. It is assumed that more fatigue fingers result in smaller angle q and velocity $\dot{q}$ [20], [21]. Note that there was no additional mechanism for physically supporting subjects' upper-limbs in experiments, such that their fingers were subjected to fatigues in exercises. The experiments were performed to not only quantify fatigue-induced interactive impedance, but also identify the functions $\overline{q} = f_1(\overline{k_p})$ and $\overline{\dot{q}} = f_2(\overline{k_d})$ given in Eq. (6). Note that this quantification and identification are numerical, rather than binary (fatigue and non-fatigue).

A. Online Mechanical Impedance Adaptation

We can see that the RAN controller enables the TIE-EXO to produce variable resistant torques in finger exercises. The torques result from online adapting mechanical impedance parameters k_p and k_d (see rows 1-3 of Fig. 5 (A)). The impedance increases with the rising absolute values (i.e., $|q|$ and $|\dot{q}|$) of the TIE-EXO angle and velocity. As a result, the online impedance adaptation leads to smooth exercise motions (see row 4 of Fig. 5 (A)). Interestingly, such online adaptation results in the convergence within a small range of average mechanical impedance parameters $\overline{k_p}(t)$ and $\overline{k_d}(t)$ (see Fig. 5 (B)). Here $\overline{k_p}(t)$ and $\overline{k_d}(t)$ are given by,

$$\overline{k_{p,d}}(t) = \frac{\overline{k_{p,d}}(t-\Delta t)(n-1) + k_{p,d}(t)}{n}, \Delta t \approx 0.06(s), \tag{7}$$

where $\overline{k_p}(t)$ and $\overline{k_d}(t)$ are the moving averages of mechanical impedance parameters $k_p(t)$ and $k_d(t)$ [1]. Δt is the time step, while n is the step length.

The RAN control exhibits impedance control variability and invariance in interaction control (see Fig. 5 (B)). This variability and invariance can be used to not only enhance exoskeleton performance (e.g., adaptivity), but also quantify fatigue-induced mechanical impedance with respect to different subjects and fingers (see Fig. 6).

B. Mechanical Impedance Quantification

$\overline{k_p}(t)$ and $\overline{k_d}(t)$ (see Eq. (7)) were used to quantify average mechanical impedance with respect to different subjects and fingers [1]. We can see that average mechanical impedance parameters decreased from 3rd, 6th, to 9th trials. This is because index and middle fingers were subjected to more fatigues in latter trials. Additionally, the male subjects (i.e., S1 and S2) exercised larger impedance resistance (see rows 1-2 of Fig. 6), compared to those performed by the female (i.e., S3). The subject S1 has a larger average mechanical impedance variance between index and middle fingers, compared to those of the subjects S2 and S3 (see columns 1-3 of Fig. 6).

We can see that variable mechanical impedance patterns can be quantified by the average mechanical impedance

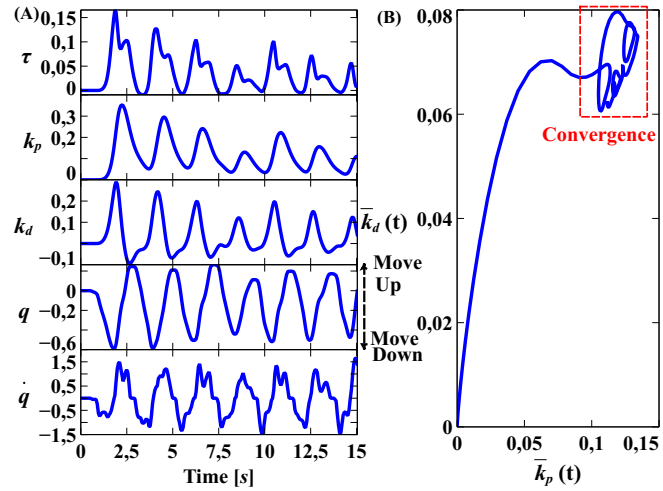

Fig. 5. Example of online mechanical adaptation by resist-as-needed (RAN) control (see Fig. 4). The data was obtained from 6th trial where the subject S1 wore the TIE-EXO to exercise his index finger. (A) Online adapted resistant torque τ, mechanical impedance parameters k_p and k_d, TIE-EXO angle q and velocity $\dot{q}$. (B) Average mechanical impedance parameters $\overline{k_p}(t)$ and $\overline{k_d}(t)$ (see Eq. (7)).

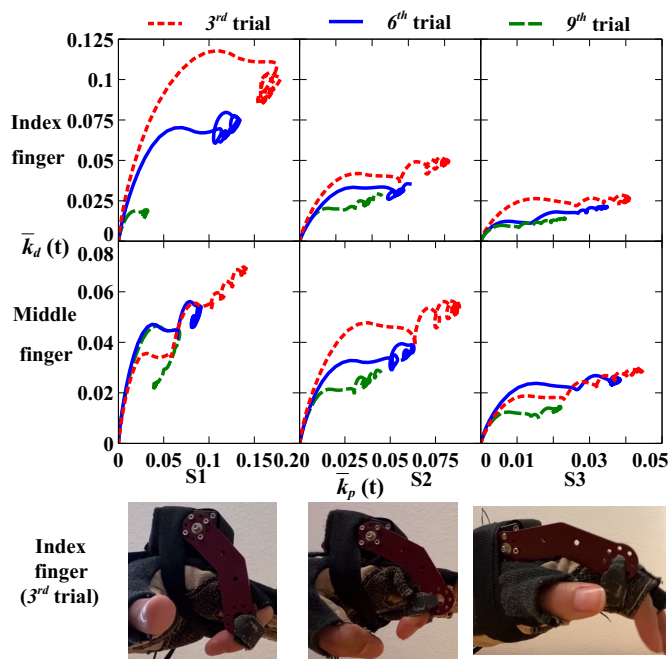

Fig. 6. Mechanical impedance qualification for different subjects and fingers in trials. The average mechanical impedance parameters $\overline{k_p}(t)$ and $\overline{k_d}(t)$ are used to quantify fingers' movements that were subjected to fatigues over trials. Columns 1-3 illustrate mechanical impedance adaptations to three subjects, while rows 1-2 depict online resistance adaptations to their index and middle finger exercises. The experimental snapshots of 3rd trials on subjects S1-3 are shown in row 3.

parameters $\overline{k_p}(t)$ and $\overline{k_d}(t)$. Additionally, the RAN controller enables the TIE-EXO to produce online adaptive resistance for exercises of different subjects and fingers. Such resist-as-needed adaptation can free physiotherapists from manually tuning mechanical impedance parameters for various subjects and fingers, that is usually required by classical controllers

[7], [8], [9], [10]. As a result, our online mechanical impedance adaptation (i.e., RAN) is more practical and effective than manually tuned variable impedance control in the exercise performance and control stability [14], [15], [22].

C. Fatigue Function Identification

The functions $f_1(\overline{k_p})$ and $f_2(\overline{k_d})$ are identified and given by (see Eq. (6)),

$$\overline{q} = f_1(\overline{k_p}), \dot{\overline{q}} = f_2(\overline{k_d}) \tag{8}$$

where $\overline{k_p}$ and $\overline{k_d}$ are the average mechanical impedance parameters for each trial with a trial time of 15 s (see Eq. (7)).

The identified data (see the red crosses in Fig. 7) is fit to the functions $f_1(\overline{k_p})$ and $f_2(\overline{k_d})$ based on a chi-squared method. The fit function $f_1(\overline{k_p})$ is given by (see Fig. 7 (A)),

$$\overline{q} = f_1(\overline{k_p}) \approx 0.069 + 0.53\overline{k_p}^{\frac{1}{2}}. \tag{9}$$

Besides, the fit function $f_2(\overline{k_d})$ is given by (see Fig. 7 (B)),

$$\dot{\overline{q}} = f_2(\overline{k_d}) \approx 0.3 + 0.65\overline{k_d}^{\frac{1}{4}}. \tag{10}$$

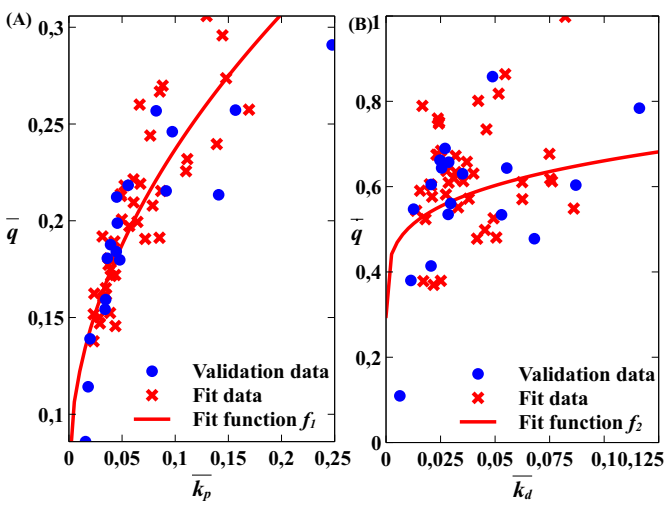

Fig. 7. Fit functions $f(\overline{k_p})$ and $f(\overline{k_d})$ (see Eqs. (9) and (10)) based on fit and validated data.

We can see that the stiffness fit function (i.e., f_1) has higher fit and validation accuracies (i.e., the lower average errors δ_1, see TABLE III), compared to those of the damping fit function (i.e., f_2). The fit and validation performance can be enhanced by increasing the amounts of trials and subjects in future work.

TABLE III

AVERAGE FIT AND VALIDATION ERRORS $\delta_{(1,2)}$ (EQ. (11))

Functions	Stiffness f_1 (Eq. (9))	Damping f_2 (Eq. (10))
Fit	0.08	0.16
Validation	0.12	0.33

The average fit and validation errors are given by,

$$\begin{aligned} \delta_1 &= \frac{\sum_1^n \frac{|\overline{q}^{(i)} - \hat{\overline{q}}^{(i)}|}{|\overline{q}^{(i)}|}}{n}, \hat{\overline{q}}^{(i)} = f_1(\overline{k_p}^{(i)}), \\ \delta_2 &= \frac{\sum_1^n \frac{|\dot{\overline{q}}^{(i)} - \hat{\dot{\overline{q}}}^{(i)})|}{|\dot{\overline{q}}^{(i)}|}}{n}, \hat{\dot{\overline{q}}}^{(i)} = f_2(\overline{k_d}^{(i)}), \end{aligned} \tag{11}$$

where $(1,2)$ denote the ID of the fit functions $f_{(1,2)}$ for the predicted average position $\hat{\overline{q}}^{(i)}$ and velocity $\hat{\dot{\overline{q}}}^{(i)}$. i is the data ID of fit or validation data with the length n. The lengths of fit and validation data are 42 and 18 (see the red crosses and blue dots in Fig. 7), respectively.

IV. Discussion and Conclusion

In this paper we present a simple and novel method for not only identifying the functions between mechanical impedance and finger fatigues (see Fig. 7), but also quantifying fatigue-induced mechanical impedance (see Fig. 6). Interestingly, this quantification indicates that there is an interactive impedance variability and invariance in fingers' movement (Fig. 5). Moreover, the RAN control enables the TIE-EXO to produce online resistance adaptations to different finger fatigues and wearers. Online impedance adaptation (i.e., RAN) control has been proven better than manually tuned impedance control in the exercise performance and control stability. More comparison details can be seen in our previous papers [14], [15].

The presented impedance quantification and adaptations have not been realized by other methods, in which the mechanical impedance (e.g., stiffness) parameters were usually kept fixed. For example, Lambercy et al. (2015) proposed various stiffness and damping constants of a virtual object for quantifying mechanical impedance of finger grip in physical manipulation tasks (i.e., grasp) [23]. Similar manual tuning methods had also been used in [7], [8], [9], [10]. However, such tuned impedance constants are not physiologically plausible, because humans tend to adapt their fingers' impedance to different manipulation tasks [11]. The implausibility may fail to adapt to finger dynamics [1]. For impedance control, most existing methods focus on assist-as-needed (ANN) control. A general review of control strategies for upper limb exoskeletons is given in [22]. To the best of our knowledge, the presented resist-as-needed control has not yet been proposed by others. In future, the proposed method can be extended in:

- Optimizing the TIE-EXO hardware design that can be applied to thumbs and little fingers.
- Integrating EMG and force sensors into the TIE-EXO. The multiple sensing can provide more accurate quantifications of mechanical impedance, compared to the presented one.

Acknowledgments

The authors would like to thank the subjects that took part in the presented experiments.

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